

Scalability & Future Plan

Date	29/09/2026
Team ID	UNM1425COMICCRAFT
Project Name	COMICCRAFT
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Runs only on a local server and is not deployed on the cloud	Cannot handle many users at the same time	High
2	Stable Diffusion image generation is slow on local hardware	Users wait longer for each comic to finish	High
3	No database or user accounts to save created comics	Users cannot save or revisit earlier comics	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Runs on a local Uvicorn server	Deploy on a cloud platform with multiple workers and a load balancer
2	Data Storage	Images and PDFs saved in static folders	Move files to cloud storage and add a database to store user comics
3	Performance	Panels generated one after another	Use a GPU and generate panel images in parallel or as background tasks
4	Security	API keys stored in a .env file	Add user login, stronger input validation and a secret manager for keys

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Deploy on the cloud and add user accounts with a saved comic library	Next 3 months	Supports more users and lets them save comics
Phase 3	Use GPU-based parallel image generation and add more art styles	3 - 6 months	Faster comic creation with more variety
Phase 4	Add multi-page story arcs, comic sharing and a mobile-friendly design	6 - 12 months	Reaches a wider audience with richer stories